NEX ENTERTAINMENT

Fullstack engineer technical test

Part A. Candidate brief

Welcome

Thanks for taking the time. This test mirrors the work you would do in your first month: Firestore rules, Cloud Functions, a React admin screen, and CSV handling. It is small on purpose. We care more about correctness and clear thinking than about how much you ship.

The scenario

Nex runs hotel campaigns with TikTok creators. A creator joins a campaign and submits a TikTok video link. Ops review each link and approve or reject it. At month end, we export the approved submissions as a payout CSV.

You are building the thin slice that handles submission, review and export.

Ground rules

- Use the Firebase emulators only (Auth, Firestore, Functions). Do not create a real Firebase project and do not commit any secrets.
- Latest Node, React 19 with TypeScript and Vite. Any styling approach is fine.
- Use the supplied seed.json (2 campaigns, 3 users, 6 submissions) and creators-messy.csv. Both are in the appendix.
- AI tools are allowed. In the follow-up call you must be able to explain every line you submit, and change it live.
- Time-box yourself to about 3 to 4 hours. If you run out of time, stub the task and say so in NOTES.md. A clear stub beats a silent gap.

Tasks

#	Task	Time	Skills
1	**Data model and security rules.** Model campaigns, submissions and auditLogs. Write firestore.rules so that a creator can read and create only their own submissions and can never set status, and ops can read everything. All other writes go through functions. Deny everything else. Add a rules test with @firebase/rules-unit-testing that shows one allowed and two denied requests.	45 min	Rules, least privilege
2	**Review callable.** Build reviewSubmission({ id, decision, reason }). Only users with the ops custom claim may call it. Allowed moves: pending to approved, pending to rejected. A rejection needs a reason. Build a second callable, markPaid, for approved to paid. Calls must be safe to repeat and safe when two people call at once. Every change writes an auditLogs entry. Add unit tests.	60 min	Functions, transactions, state machine
3	**React screens.** Ops: a review queue with filters for status and campaign, and approve and reject actions, with loading, empty and error states. Creator: a form that submits a TikTok URL with validation, plus a list of their own submissions.	60 min	React, forms, UX states

#	Task	Time	Skills
4	**CSV.** (a) Import creators-messy.csv. Report each bad row with its line number and a reason, and keep importing the good rows. (b) Export approved submissions for a chosen month as a payout CSV that opens correctly in Excel, with the amount taken from the campaign rate.	30 min	Parsing, encoding, escaping
5	**Language toggle.** The UI switches between English and Thai. Add a check (script or test) that fails when a key exists in one language but not the other.	15 min	i18n
6	**NOTES.md, one page.** (a) How would you deploy only your new function, without touching other functions? (b) Where would a TikTok client secret live? (c) What happens if two ops approve the same submission at the same moment in your design? (d) Two things you would fix with more time.	10 min	Deploy safety, judgement

What to send back

- A git repository (or zip) with a README that has the exact commands to start the emulators, run the app and run the tests.
- **A screen-recorded video, 5 to 8 minutes, of you explaining your app.** Show it running, then walk us through your code. Details below.
- NOTES.md with your answers to task 6, plus anything you stubbed or cut.
- Commit as you go. We read the history.

The video walkthrough

We want to hear you think. Record your screen and your voice (your face is optional). Use Loom, QuickTime, OBS or any tool you like. Upload it as an unlisted link or attach the file, with a maximum of 8 minutes. Do not read from a script, and do not worry about polish.

Please cover these four things, in this order:

1. Demo (about 2 min). Show the creator submitting a link, ops approving and rejecting it, and the CSV export. Show one thing that fails on purpose, such as a denied request or a bad CSV row, and what the user sees.
2. Rules and callable (about 2 min). Walk through your firestore.rules create rule and your reviewSubmission function. Explain what stops a creator from approving their own submission.
3. A decision you made (about 2 min). Pick one choice where you had options, say what you chose and why, and what you gave up.
4. What is missing (about 1 min). Tell us what you stubbed, what you would fix first, and what worries you most.

What we look at

Security of the rules and callables, correctness under repeated and concurrent calls, how the screens behave when things go wrong, how you handle messy data, and how clearly you explain your choices in the video.

Appendix. Seed data

The same two files are saved next to this document as seed.json and creators-messy.csv.

seed.json


```
{
  "users": [
    {
      "uid": "creator-a",
      "email": "a@example.com",
      "role": "creator"
    },
    {
      "uid": "creator-b",
      "email": "b@example.com",
      "role": "creator"
    },
    {
      "uid": "ops-1",
      "email": "ops@example.com",
      "role": "ops"
    }
  ],
  "campaigns": [
    {
      "id": "camp-bkk",
      "title": "Bangkok Riverside Hotel",
      "status": "open",
      "ratePerVideoTHB": 500
    },
    {
      "id": "camp-pkt",
      "title": "Phuket Beach Resort",
      "status": "open",
      "ratePerVideoTHB": 800
    }
  ],
  "submissions": [
    {
      "id": "s1",
      "campaignId": "camp-bkk",
      "creatorId": "creator-a",
      "tiktokUrl": "https://www.tiktok.com/@a/video/7300000000000000001",
      "status": "pending",
      "createdAt": "2026-09-10T08:00:00Z"
    },
    {
      "id": "s2",
      "campaignId": "camp-bkk",
      "creatorId": "creator-b",
      "tiktokUrl": "https://www.tiktok.com/@b/video/7300000000000000002",
      "status": "pending",
      "createdAt": "2026-09-12T10:30:00Z"
    },
    {
      "id": "s3",
      "campaignId": "camp-pkt",
      "creatorId": "creator-a",
      "tiktokUrl": "https://vm.tiktok.com/ZSabc123/",
      "status": "approved",
      "createdAt": "2026-09-15T04:00:00Z"
    },
    {
      "id": "s4",
      "campaignId": "camp-pkt",
      "creatorId": "creator-b",
```

```
      "tiktokUrl": "https://www.tiktok.com/@b/video/7300000000000000004",
      "status": "approved",
      "createdAt": "2026-09-30T18:30:00Z"
    },
    {
      "id": "s5",
      "campaignId": "camp-bkk",
      "creatorId": "creator-a",
      "tiktokUrl": "https://www.tiktok.com/@a/video/7300000000000000005",
      "status": "rejected",
      "reason": "Hotel not visible",
      "createdAt": "2026-09-20T09:00:00Z"
    },
    {
      "id": "s6",
      "campaignId": "camp-pkt",
      "creatorId": "creator-a",
      "tiktokUrl": "https://www.tiktok.com/@a/video/7300000000000000006",
      "status": "paid",
      "createdAt": "2026-08-28T09:00:00Z"
    }
  ]
}
```


creators-messy.csv


```
name,email,tiktok_handle,city
Somchai Jaidee,somchai@example.com,@somchai.travel,Bangkok
"Nguyen, Linh",linh@example.com,@linhtrips,"Chiang Mai, TH"
สมศรี ใจดี,somsri@example.com,@somsri_th,Phuket
Duplicate Person,somchai@example.com,@dup_handle,Bangkok
,not-an-email,@broken,Pattaya
=HYPERLINK("http://evil.example"),formula@example.com,@formula,Krabi
Mali Wong,mali@example.com,mali.no.at,Bangkok
```